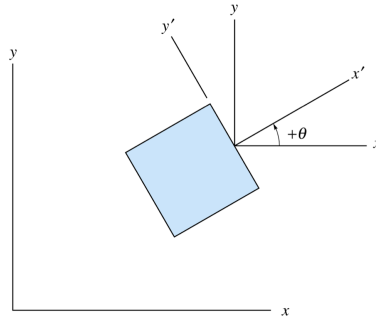
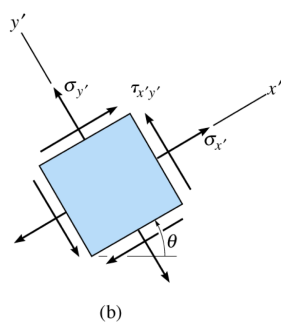
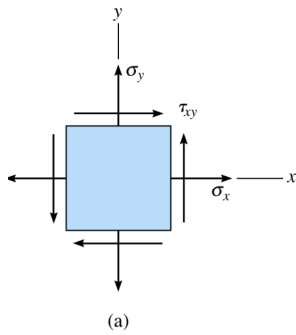


Stress Transformation



逆时针, $\theta > 0$

$$\sigma_{y'} = \sigma_{x'}(\theta + \frac{\pi}{2})$$

$$\sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos(2\theta) + \tau_{xy} \sin(2\theta)$$

$$\sigma_{y'} = \frac{\sigma_x + \sigma_y}{2} - \frac{\sigma_x - \sigma_y}{2} \cos(2\theta) - \tau_{xy} \sin(2\theta)$$

$$\tau_{x'y'} = -\frac{\sigma_x - \sigma_y}{2} \sin(2\theta) + \tau_{xy} \cos(2\theta)$$

Note: $\text{Trace} \begin{bmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_y & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_z \end{bmatrix} = \sigma_x + \sigma_y + \sigma_z$ is const.

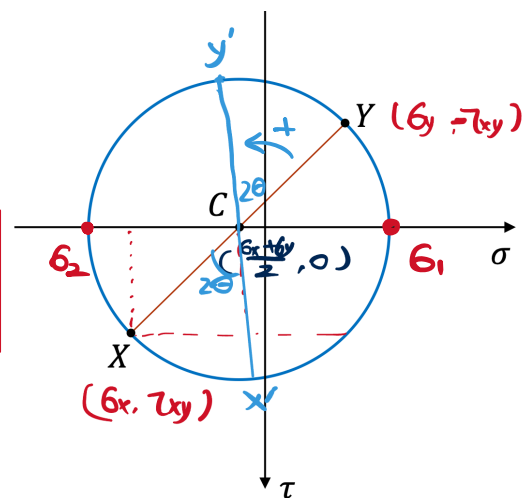
* 有时给的角未必是 θ , 先确定 θ !!!

Mohr's Circle in 2D

$X(\sigma_x, \tau_{xy})$ $Y(\sigma_y, -\tau_{xy})$

* 转 θ 度. 圆内 $x \rightarrow y'$ 相对 $x-y$ 转 2θ 度

$$\sigma_c = \frac{\sigma_x + \sigma_y}{2}, \quad R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$



Principle Stresses

First Principle σ : $\sigma_{\max} = \sigma_1 = \sigma_c + R$, $\tan(2\theta_{p1}) = \frac{2\tau_{xy}}{\sigma_x - \sigma_y}$

Second Principle σ : $\sigma_{\min} = \sigma_2 = \sigma_c - R$, $\theta_{p2} = \theta_{p1} + \frac{\pi}{2}$

No shear stress: $\tau_{x'y'}(\theta_{p1}/\theta_{p2}) = 0$

Maximum Shear Stress

$$|\tau_{\max}| = R$$

$$\tan(2\theta_s) = \frac{-(\sigma_x - \sigma_y)}{2\tau_{xy}}$$

$$\text{when } \sigma_x' = \sigma_y' = C$$

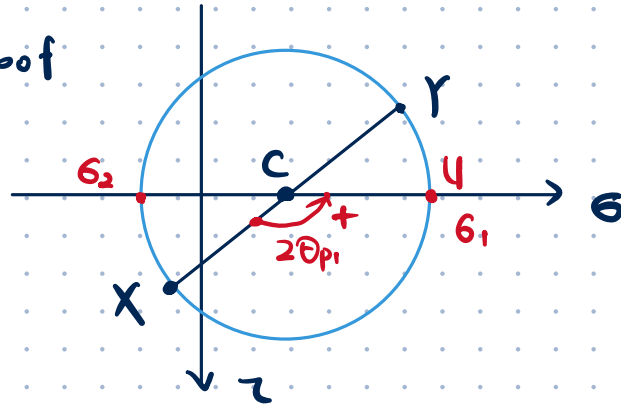
$$\theta_s = \theta_p \pm \frac{\pi}{4}$$

$$\theta_{s1} + \frac{\pi}{2} = \theta_{s2} \quad 2 \text{ roots}$$

* θ_{p1} calculation proof

$$X(\sigma_x, \tau_{xy})$$

$$Y(\sigma_y, -\tau_{xy})$$



将 CX 转至 σ_1 正方向 U
的角度为 $2\theta_{p1}$

$$U(\sigma_c + R, 0)$$

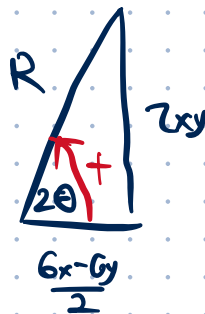
$$C(\sigma_c, 0)$$

$$\sigma_c = \frac{\sigma_x + \sigma_y}{2}$$

$$\vec{CU} = (R, 0)$$

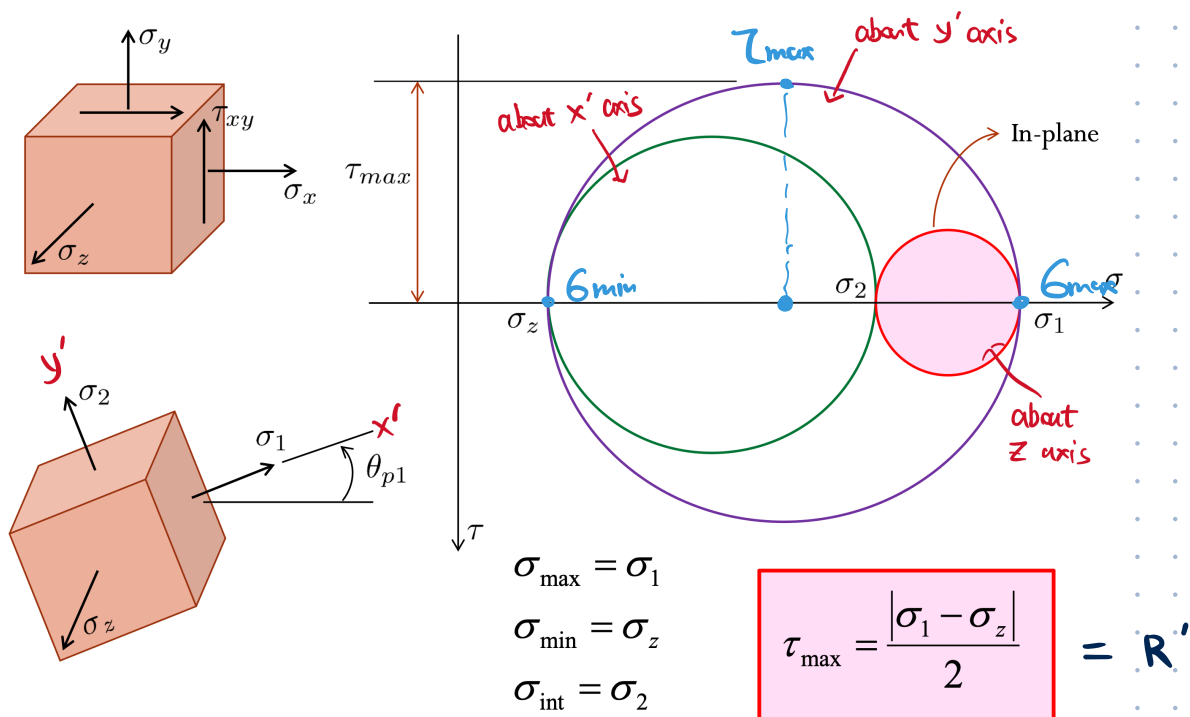
$$\vec{CX} = \left(\frac{\sigma_x - \sigma_y}{2}, \tau_{xy}\right)$$

$$\cos(2\theta) = \frac{\vec{CU} \cdot \vec{CX}}{|\vec{CU}| |\vec{CX}|} = \frac{R \frac{\sigma_x - \sigma_y}{2}}{R \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}}$$



$$\Rightarrow \tan(2\theta_{p1}) = \frac{2\tau_{xy}}{\sigma_x - \sigma_y}$$

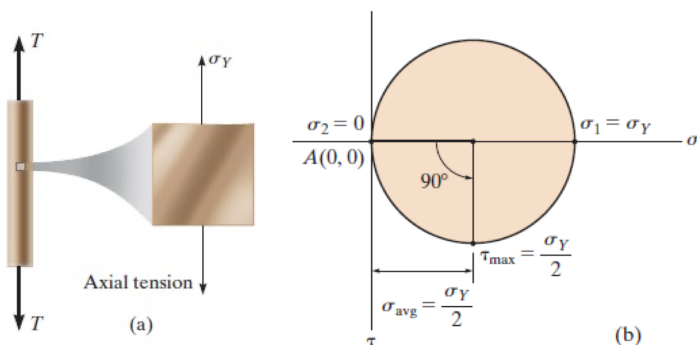
3D state of stress



Theories of Failure

Tresca Criteria (maximum - shear - stress criterion) * consider 3D case

Typically, Ductile material fail by shear stresses



← In a Test to get G_r

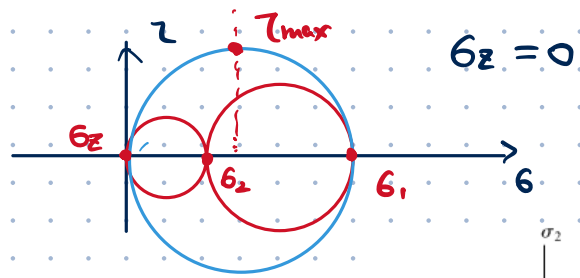
$$\tau_{max} = R_T = \frac{G_r}{2}$$

* 若 F/T 未知, 则用
以下所有情况算 F/T 最小值

① σ_1, σ_2 同号

$$\tau_{max} = \frac{|\sigma_1|}{2} \text{ or } \frac{|\sigma_2|}{2}$$

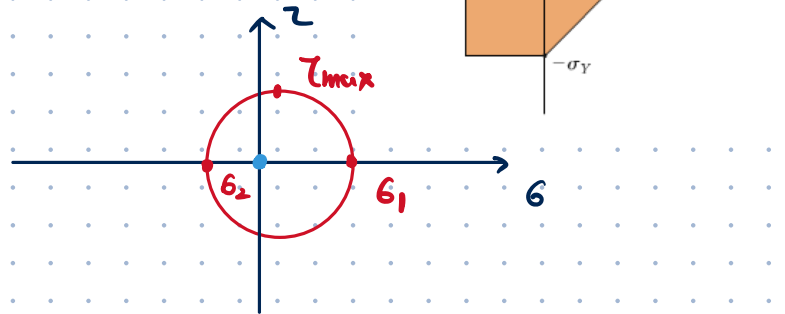
$$\sigma_r \geq |\sigma_1| \text{ or } |\sigma_2|$$



② σ_1, σ_2 异号

$$\tau_{max} = \frac{|\sigma_1 - \sigma_2|}{2}$$

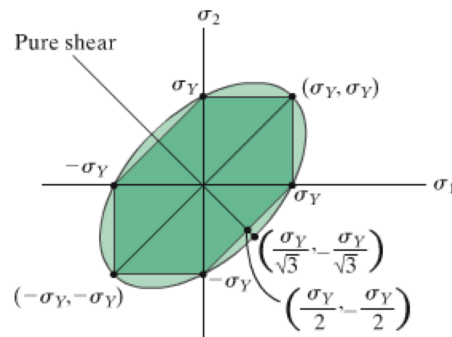
$$\sigma_r \geq |\sigma_1 - \sigma_2| = 2\tau$$



Von-Mises Criterion (maximum-distortion-energy criterion)

Applied for ductile materials

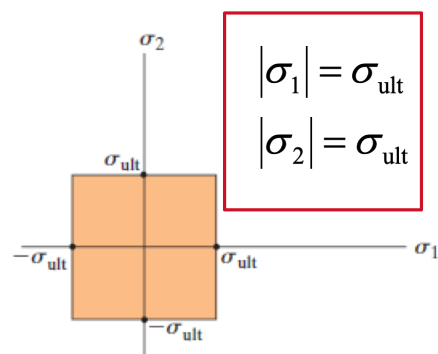
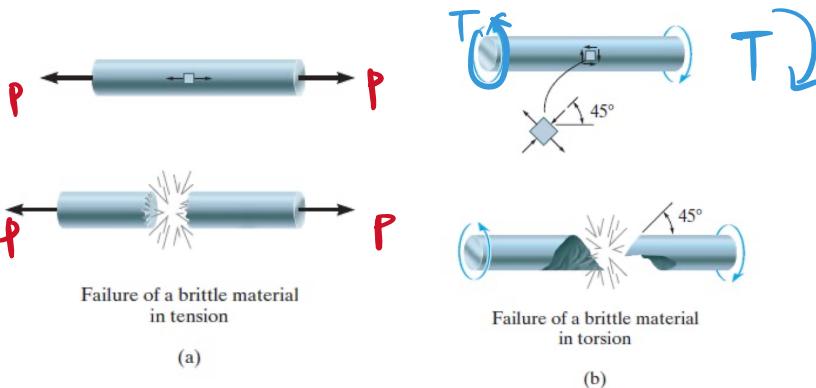
$$\sigma_1^2 - \sigma_1\sigma_2 + \sigma_2^2 = \sigma_r^2$$



Maximum-Normal-Stress Criterion

Applied for Brittle Materials

* Due to Imperfection, 未必可用



Failure Surfaces

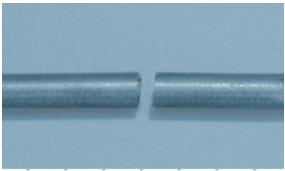
Brittle Material

Tension

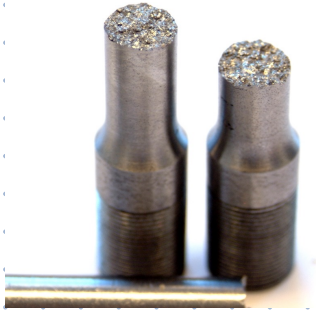
垂直切面

Normal Stress

σ_1, σ_2



Failure of a brittle material in tension



Torsion

切面 45°

Normal Stress

σ_1, σ_2



Failure of a brittle material in torsion

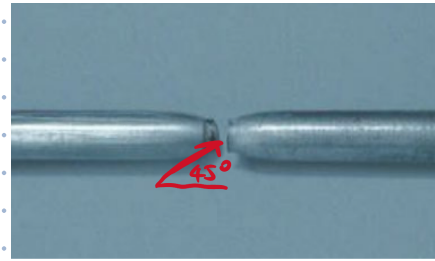
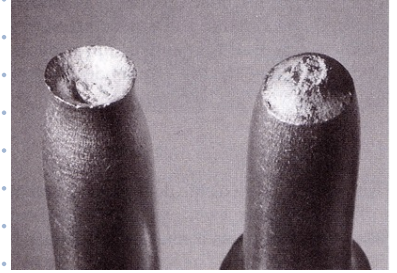
Ductile Material

Tension

45° 拉伸面

Shear Stress

τ_{max}

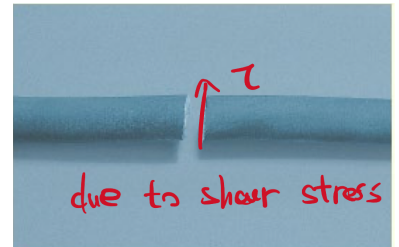
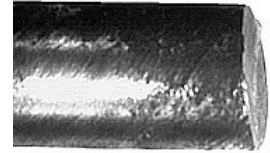


Torsion

垂直切面

Shear Stress

τ_{max}



due to shear stress